SQL COMMANDS COMPARISON GUIDE

PostgreSQL vs MySQL Detailed Reference Manual

Yeh PDF aapke references (**"mujhe ek sepereate list bnakr dijiye postgresql k.pdf"** aur **"SQL COMMANDS .pdf"**) ke data ko organize karke banayi gayi hai. Isme Common Commands aur Dono ke beech ke major Differences ko proper tabular aur structural layout me samjhaya gaya hai taaki aap ise as a quick reference use kar sakein.

1. Overview & Comparison Quick Summary

Relational Databases (RDBMS) hone ke naate, PostgreSQL aur MySQL dono hi SQL standard ko tightly follow karte hain. Is wajah se basic data extraction aur modifications dono me identical hote hain. Halanki, enterprise architecture, syntax strictness, terminal commands, aur advanced data manipulation capabilities ke mamle me dono me mahatvapurna antar hain.

FEATURE / CONCEPT	POSTGRESQL SYNTAX & BEHAVIOR	MYSQL SYNTAX & BEHAVIOR	STATUS
Auto-Increment IDs	`id SERIAL PRIMARY KEY` Parde ke peeche automatic sequence generator banata hai.	`id INT AUTO_INCREMENT PRIMARY KEY` Explicit keyword ka use karta hai auto-generation ke liye.	DIFFERENT
String Literal Quotes	Strictly single quotes `'Sahil'` Double quotes ka use strict identifiers (tables/columns) ke liye hota hai.	Dono single aur double quotes support karta hai: `'Sahil' ya "Sahil"`	DIFFERENT
Full Outer Join	Native Support: `SELECT * FROM t1 FULL OUTER JOIN t2 ON t1.id = t2.id`	Native support nahi hai. Iske liye Left aur Right Join ka `UNION` workaround lagana padta hai.	DIFFERENT

FEATURE / CONCEPT	POSTGRESQL SYNTAX & BEHAVIOR	MYSQL SYNTAX & BEHAVIOR	STATUS
Text Search Case-Sensitivity	`LIKE` case-sensitive hota hai. Case-insensitive ke liye explicit `ILIKE` keyword ka use hota hai.	Default `LIKE` query standard collations (jaise utf8mb4_general_ci) me case-insensitive hi hoti hai.	DIFFERENT
CRUD & Joins Core	Standard ANSI SQL syntax. `SELECT, INSERT, UPDATE, DELETE`	Standard ANSI SQL syntax. `SELECT, INSERT, UPDATE, DELETE`	COMMON

2. Terminal & Meta Commands (CLI Differences)

PostgreSQL apne psql terminal utility ke shortcut codes use karta hai jabki MySQL standard SQL logs / actions commands engine par chalta hai:

ACTION DESCRIPTION	POSTGRESQL TERMINAL COMMAND	MYSQL UTILITY COMMAND
Saare databases ki list dekhna	`\l`	`SHOW DATABASES;`
Kisi database se connect / switch karna	`\c database_name`	`USE database_name;`
Database ke tables ki list dekhna	`\dt`	`SHOW TABLES;`
Table ka schema / structure dekhna	`\d table_name`	`DESCRIBE table_name;` `ya EXPLAIN table_name;`
Terminal session se exit karna	`\q`	`EXIT; ya QUIT;`

3. Common Commands Reference (100% Identical Syntax)

Yeh commands dono databases me bina kisi badlav ke execute hoti hain. Dono systems in standard operations ke liye identical functionality output karte hain:

Data Definition (DDL) & Constraints

- **Database Operations:** `CREATE DATABASE db_name;` aur `DROP DATABASE db_name;`
- **Table Operations:** `DROP TABLE table_name;` aur `TRUNCATE TABLE table_name;` (Table empty karne ke liye).
- **Standard Constraints:** Table creation ke waqt dono systems `NOT NULL`, `UNIQUE`, `PRIMARY KEY`, aur `FOREIGN KEY` constraints ko standard format me apply karte hain.

Data Manipulation Language (DML)

Core CRUD syntax bilkul standard hai:

```
-- Insert Data
INSERT INTO students (sid, sname, email) VALUES (1, 'Sahil', 'sahil@gmail.com');

-- Update Data
UPDATE students SET sname = 'Sahil Kumar' WHERE sid = 1;

-- Delete Data
DELETE FROM students WHERE sid = 1;
```

Data Querying, Filtering & Aggregation

- **Sorting:** Dono databases `ORDER BY column_name DESC/ASC` support karte hain.
- **Aggregation:** Grouping ke liye `GROUP BY column_name HAVING COUNT(*) > X` ka sequential structure bilkul same hai.
- **Mathematical & String Functions:** Core functions jaise `ABS()`, `ROUND()`, `CEIL()`, `FLOOR()`, `UPPER()`, `LOWER()`, aur `LENGTH()` dono me valid hain.

4. Advanced Differences & Query Structure

A. Altering Columns (Table Structure Modify Karte Waqt)

Kisi specific column ka datatype change karne ka tarika dono me alag hai:

MySQL Syntax:

```
ALTER TABLE students MODIFY COLUMN sname INT;
```

PostgreSQL Syntax:

```
ALTER TABLE students ALTER COLUMN sname TYPE INT;
```

B. Full Outer Join (Native vs Simulation)

PostgreSQL full tables merger support karta hai jabki MySQL me hume set theory use karni padti hai:

PostgreSQL Direct Query:

```
SELECT * FROM membership m FULL OUTER JOIN users u ON m.uid = u.id;
```

MySQL Workaround Query (Jaisa SQL Commands manual me hai):

```
SELECT * FROM membership m LEFT OUTER JOIN users u ON m.uid = u.id
UNION
SELECT * FROM membership m RIGHT OUTER JOIN users u ON m.uid = u.id;
```

C. Performance & Explain Analyze

PostgreSQL me query performance deep profiling ke liye execution path ke sath exact run-time time report dekhne ke liye command use ki jaati hai:

```
EXPLAIN ANALYZE SELECT * FROM users WHERE email = 'sahil@example.com';
```

MySQL me query plan dekhne ke liye simple `EXPLAIN` ya updated versions me `EXPLAIN ANALYZE` use hota hai par graphical aur processing trace flow internals ka kafi difference hota hai.

SQL Comparison Technical Guide • Generated for reference tracking database execution modules.